Multimodal AI for Early Detection and Interpretation of Cardiac Disorders Using ECG and MRI Data

ECG Signal Analysis

1. What is an ECG?

An ECG (electrocardiogram) records the electrical activity of the heart over time. Each heartbeat produces a repeating pattern of waves. The ECG captures and traces this activity as a continuous curve.

1.1 How Are Signals Recorded?

Electrodes are placed on the skin (arms, legs, chest). Each pair of electrodes forms a lead that captures the signal from a different angle.

- A standard ECG has 12 leads.
- 12 points of view on the same heart, at the same time.

1.2 Sampling Frequency

Sampling frequency is the number of measurements of the electrical signal taken per second.

- Higher frequency → more temporal detail.

- Higher frequency → heavier data (larger file size).

ECG at 500 Hz: 500 measurements per second. Each of the 12 leads independently records the heart's electrical voltage 500 times in 1 second, all at the same time points.

1.3 Data Dimensions (Standard ECG)

Item	Value
Sampling frequency	500 Hz
Typical recording duration	10 seconds
Points per lead	500 × 10 = 5,000 points
Number of leads	12
Total values per patient	5,000 × 12 = 60,000 values

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2. How to Read an ECG

Every heartbeat follows a consistent sequence of electrical waves:

P → QRS → T

- In a healthy person, this cycle lasts about 0.6–1.0 seconds.
- The order and timing of these waves are identical across all 12 ECG leads.
- What changes from lead to lead is the appearance, shape, size, and direction.
- Each lead captures the same electrical event from a different angle.

2.1 The Waves

P Wave

- Represents: Atrial contraction (upper chambers push blood into

ventricles).
- Normal duration: ~80–120 ms
- Abnormal if: Absent → atrial problem (e.g., atrial fibrillation)

QRS Complex

- Represents: Ventricular contraction, the actual heartbeat. Most visible peak.
- Normal duration: ~80–100 ms
- Abnormal if: Wide and deformed → bundle branch block or ventricular arrhythmia

T Wave

- Represents: Ventricular repolarization (ventricles recharge electrically).
- Abnormal if: Inverted or flattened → sign of ischemia (lack of oxygen to the heart)

2.2 The Intervals

RR Interval

- Definition: Distance between two consecutive R peaks
- Gives: Heart rate
- Abnormal if: Irregular → arrhythmia

PR Interval

- Definition: From the start of P to the start of QRS
- Measures: Atrial → ventricular conduction time
- Abnormal if: Too long → AV block

ST Segment

- Definition: The line between the end of QRS and the start of T
- Should be: On the baseline (isoelectric)
- Abnormal if: Elevated or depressed → possible myocardial infarction (heart attack)

QT Interval

- Definition: From the start of QRS to the end of T
- Abnormal if: Too long → risk of serious arrhythmia (e.g., Torsade de Pointes)

2.3 Reference Table: Waves & Intervals

Wave / Interval	What It Shows	Normal	Abnormal =
P wave	Atrial contraction	~80–120 ms	Absent → AFib
QRS complex	Ventricular contraction	~80–100 ms	Wide → block/ arrhythmia
T wave	Ventricular recharge	Upright	Inverted → ischemia
RR interval	Heart rate	Regular	Irregular → arrhythmia
PR interval	AV conduction	~120–200 ms	Too long → AV block
ST segment	Between depolarization & repolarization	Flat	Elevated/ depressed → MI
QT interval	Total ventricular activity	<440–460 ms	Too long → Torsade de Pointes

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3. Key Disorders for Classification

3.1 Atrial Fibrillation (AF)

A heart rhythm disorder where the atria no longer contract in an organized way: irregularity detection in beat sequences.

- ECG pattern: Absent P waves + irregularly irregular RR intervals

- Clinical note: Most common arrhythmia

3.2 ST-Elevation Myocardial Infarction (STEMI)

A myocardial infarction (heart attack) with ST segment elevation.

- ECG pattern: ST elevation ≥1 mm in ≥2 contiguous leads + reciprocal depression

Note: Elevation = the signal goes ABOVE the line. Depression = the signal goes BELOW the line. When both are seen simultaneously, elevation in some leads and depression in others, this confirms a real MI from two angles. The location of elevation and depression follows a precise anatomical logic.

The heart has 4 walls. Each wall is "watched" by specific leads:

Wall	Leads watching it	Reciprocal leads (opposite)
Inferior (bottom)	II, III, aVF	I, aVL
Anterior (front)	V1, V2, V3, V4	None direct
Lateral (left side)	I, aVL, V5, V6	II, III, aVF
Posterior (back)	V7, V8, V9	V1, V2, V3

- Clinical note: Time-critical emergency

3.3 Bundle Branch Blocks (LBBB / RBBB)

A bundle branch block is a delay or interruption in the electrical conduction through the ventricles. The electrical impulse originates in the atria, passes through the AV node, then propagates rapidly into the ventricles via two branches (right and left).

- ECG pattern: Wide QRS >120 ms + specific lead patterns:
  - **RBBB** (Right Bundle Branch Block) → typical RSR' pattern in lead V1 (two peaks)

    - **LBBB** (Left Bundle Branch Block) → typical wide QS pattern in lead V1
- Classification type: Morphology classification

3.4 Left Ventricular Hypertrophy (LVH)

Increase in the muscle mass of the left ventricle. The left ventricle must sometimes work harder (e.g., in hypertension or valvular stenosis). To compensate, its muscle thickens → it becomes more massive.

Result on ECG: R waves are much taller (greater amplitude), as a larger muscle generates a stronger electrical signal.

- ECG pattern: High-voltage R waves
- **Sokolow-Lyon criterion:** SV1 + RV5 ≥ 35 mm → LVH
- **Modified Cornell criterion:** R wave in aVL > 1.2 mV → LVH
- Classification type: Voltage measurement task

MRI Analysis

1. What is an MRI?

Magnetic Resonance Imaging (MRI) is a non-invasive medical imaging technique that uses strong magnetic fields and radio waves to create detailed images of the organs and tissues inside the body.

1.1 How Are MRIs Recorded?

The body is mostly water. Water contains hydrogen atoms. MRI uses a strong magnet to align those atoms, then fires radio waves to disturb them. As the atoms "snap back" to alignment, they emit a signal. A computer turns that signal into an image.

1.2 Why MRI for the heart specifically?

MRI is the gold standard for cardiac imaging because it can show:

- **Heart movement:** MRI films the heart beating in real time → Function
- **Scar tissue (fibrosis):** Dead heart muscle from a past heart attack shows up as a bright white patch → Tissue damage
- **Chamber size & wall thickness:** Precise measurements of the ventricles → Structure
- **Blood flow (perfusion):** Contrast agent injected → shows which areas receive enough blood → Blood supply

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2. Cardiac MRI Sequences

A "sequence" is a different type of MRI scan. Unlike ECG which always records the same signal, MRI has multiple "modes" called sequences. Each sequence highlights different tissue properties.

2.1 Cine MRI

A video of the beating heart, typically 20–30 frames per cardiac cycle. Shows how well the walls move and contract. Used to measure ejection fraction (EF).

Normal values	Pathological values	Interpretation
Ejection Fraction (EF): 55–70%	EF <55% (moderate), EF <40% (severe)	Global cardiac function (contractility)
Wall motion: normal	Hypokinetic (reduced), akinetic (absent), dyskinetic (paradoxical)	Regional wall function
Synchrony: simultaneous contraction	Dyssynchrony (delayed contraction between walls)	Mechanical coordination

2.2 LGE (Late Gadolinium Enhancement)

A contrast agent (Gadolinium) is injected. Dead/scarred tissue holds the agent longer → appears bright white. The gold standard for detecting old heart attacks.

2.3 T1 / T2 Mapping

Measures how fast hydrogen atoms relax. Different values for normal vs inflamed vs fibrotic tissue. Detects myocarditis (inflammation) and edema (swelling).

Parameter	Normal values	If High	If Low	Interpretatio
T1 native	~950–1050 ms (1.5T/3T)	Fibrosis, infiltration (e.g., amyloidosis), inflammation	Fat infiltration, iron overload (hemochromatosis)	Reflects tissue composition changes
T2	~45–55 ms	Edema, myocarditis, acute infarction	Iron overload (shortened relaxation), reduced water content	Detects acute inflammation vs iron deposition
ECV (Extracellular Volume)	~25%	Diffuse fibrosis, infiltration, extracellular matrix expansion	Fat infiltration, iron overload (reduced extracellular space)	Quantifies extracellular space abnormalities

2.4 Short Axis Stack

A series of 2D slices from base to apex of the heart. Stacked together → 3D volume of the heart.

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3. What Clinicians Measure from MRI

1. **Ejection Fraction (EF):** What % of blood the heart pumps per beat. Normal ≥55%. Below 35% = severe heart failure.
2. **End-diastolic volume (EDV):** How much blood fills the ventricle at rest. Enlarged → cardiomyopathy.
3. **Wall motion score:** Which wall segments move normally vs. poorly vs. not at all.
4. **LGE extent (%):** What percentage of the heart muscle is scarred. Predicts risk of sudden death.

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4. Cardiac Pathologies Detectable from MRI

4.1 Functional Abnormalities (from Cine MRI)

- **Heart failure (reduced EF):** The ventricle isn't pumping out enough blood per beat. HFrEF (reduced ejection fraction) directly detectable via EF; HFpEF (preserved EF) needs volumes/filling pattern.
- **Wall motion abnormalities:** Segments can be hypokinetic (weakly), akinetic (not at all), or dyskinetic (wrong direction). Sign of localized damage.
- **Dyssynchrony:** One wall contracts late relative to others → mechanically inefficient pump.

4.2 Structural Abnormalities (from Cine MRI + Short Axis Stack)

- **Dilated cardiomyopathy (DCM):** Ventricle chamber enlarged/ stretched out, walls thin. Usually accompanied by low EF.
- **Hypertrophic cardiomyopathy (HCM):** Walls abnormally thick, chamber normal or small. Often genetic. Dangerous due to obstruction and sudden cardiac events.

4.3 Tissue/Scar Abnormalities (from LGE)

- **Prior myocardial infarction:** LGE shows subendocardial (inner layer only) vs transmural (full wall thickness) patterns. Location matches coronary artery blocked.

- **Sudden death risk (via LGE extent %):** More scar tissue (% of total heart muscle) → higher risk of dangerous arrhythmias and sudden cardiac death.

4.4 Inflammatory / Infiltrative Disease (from T1/T2 Mapping)

- **Myocarditis (acute inflammation):** Elevated T2 (edema/ inflammation) and sometimes elevated T1.
- **Edema:** Excess fluid in tissue. Marker of acute injury or inflammation. Elevated T2 is the key signal.
- Amyloidosis / infiltrative disease
- Iron overload (hemochromatosis)
- Diffuse fibrosis (via ECV)